The Unified Theory of Rail and Air Transport

Infrastructure, Economics, Optimality, and Societal Equilibria under the Primary-Rail Axiom

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Abstract

We develop a unified theoretical framework for rail and air transport systems founded on a single axiom: aggregate railway traffic T_{R} is at least as large as aggregate airway traffic T_{A} at every spatial and temporal scale of aggregation. From this Primary-Rail Axiom we derive theorems in infrastructure network design, price equilibrium, urban spatial economics, environmental accounting, and sociotechnical dynamics. The framework integrates graph theory, mechanism design, general equilibrium, stochastic optimisation, and machine learning. We provide formal proofs, algorithms for network optimisation under the primary-rail constraint, econometric specifications for demand estimation, TikZ schematics of the resulting transport architectures, and comparative statistics drawn from historical and projected data. The theory is internally consistent, falsifiable, and points to a concrete policy pathway for economies that wish to satisfy the axiom.

The paper ends with "The End"

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1 Introduction

Transport systems mediate economic exchange, urban form, and social life. While the twentieth century was defined by the rapid rise of commercial aviation and the automobile—relegating rail to a residual role outside dense European and East Asian corridors—this paper reverses that presumption. We construct a coherent, formally derived theory of transport in which the inequality $T_{\mathrm{R}} \geq T_{\mathrm{A}}$ is taken as an axiomatic primitive and its implications are traced across every domain in which transport interacts with society.

The approach is novel in three respects. First, the inequality is treated not as an empirical observation but as a normative design constraint that shapes theoretical predictions in the same way that profit maximisation shapes firm theory. Second, results from operations research, urban economics, environmental science, and machine learning are synthesised into a single coherent framework rather than left in disciplinary silos. Third, every major result is accompanied by a formal proof or an operational algorithm, making the theory directly implementable.

The paper proceeds as follows. Section 2 states the foundational axiom and its mathematical apparatus. Section 3 analyses infrastructure network design. Section 4 develops the economic theory of price and quantity equilibria. Section 5 treats urban spatial economics. Section 6 addresses environmental and energy accounting. Section 7 examines social and cultural dynamics. Section 8 presents algorithms and data-science methods. Section 9 applies machine learning to demand forecasting and policy design. Section 10 concludes. A glossary of terms and notation follows the bibliography.

2 The Foundational Axiom and Mathematical Setup

2.1 Traffic measures

Definition 2.1 (Traffic measure)**.** Let $\mathcal{G} = (V, E)$ be a directed multigraph representing a transport network, where V is the set of nodes and E the set of edges. For mode $m \in \{\mathrm{R}, \mathrm{A}\}$ (rail and air), the *traffic measure* over horizon $[0, T]$ is

$$T_m = \alpha P_m + \beta F_m, \tag{1}$$

where P_m and F_m are aggregate passenger-kilometres and freight tonne-kilometres, and $\alpha, \beta > 0$ are commensuration weights.

Axiom 2.2 (Primary-Rail Axiom (PRA))**.** At every spatial scale $s \in \mathcal{S}$ and every time interval $[t_1, t_2] \subseteq [0, \infty)$,

$$\boxed{T_{\mathrm{R}}(s, t_1, t_2) \;\geq\; T_{\mathrm{A}}(s, t_1, t_2).} \tag{2}$$

Remark 2.3. The axiom must hold *simultaneously* at all scales. A configuration in which rail dominates nationally but air dominates sub-regionally violates the PRA at the sub-regional scale.

2.2 Feasibility conditions

Let $\mathcal{F}$ be the set of all physically realisable configurations $(P_{\mathrm{R}}, F_{\mathrm{R}}, P_{\mathrm{A}}, F_{\mathrm{A}})$. A configuration $\boldsymbol{x} \in \mathcal{F}$ is *PRA-feasible* if $T_{\mathrm{R}}(\boldsymbol{x}) \geq T_{\mathrm{A}}(\boldsymbol{x})$. Denote the feasible subset by $\mathcal{F}^*$.

Proposition 2.4 (Non-emptiness of $\mathcal{F}^*$)**.** *Under minimal assumptions (rail and air infrastructure exist and total demand is positive), $\mathcal{F}^*$ is non-empty.*

Proof. Assign all demand to rail and set air traffic to zero. Then $T_{\mathrm{R}} > 0$ and $T_{\mathrm{A}} = 0$, so $T_{\mathrm{R}} \geq T_{\mathrm{A}}$ holds and $\boldsymbol{x}_0 \in \mathcal{F}^*$. □

2.3 The modal split function

Define the modal split ratio

$$\mu = \frac{T_{\mathrm{R}}}{T_{\mathrm{R}} + T_{\mathrm{A}}} \in [0, 1]. \tag{3}$$

The PRA requires $\mu \geq \frac{1}{2}$. Under the cost and utility conditions derived later, the socially optimal μ^* satisfies $\mu^* > 1/2$ strictly for all continental geographies.

3 Infrastructure Network Theory

3.1 Graph-theoretic formulation

Let $\mathcal{G}_{\mathrm{R}} = (V_{\mathrm{R}}, E_{\mathrm{R}})$ and $\mathcal{G}_{\mathrm{A}} = (V_{\mathrm{A}}, E_{\mathrm{A}})$ be the rail and air graphs. The integrated transport graph is $\mathcal{G} = \mathcal{G}_{\mathrm{R}} \cup \mathcal{G}_{\mathrm{A}}$ with shared interchange nodes $V_{\mathrm{RA}} = V_{\mathrm{R}} \cap V_{\mathrm{A}}$.

Definition 3.1 (Network capacity)**.** For each edge $e \in E_m$, let $\kappa(e)$ be its capacity in transport units per unit time. The network capacity of mode m is

$$K_m = \sum_{e \in E_m} \kappa(e) \cdot \ell(e), \tag{4}$$

where $\ell(e)$ is edge length.

Theorem 3.2 (Capacity primacy)**.** *A necessary condition for the PRA to hold in expectation under Poisson demand arrivals is* $K_{\mathrm{R}} \geq K_{\mathrm{A}}$.

Proof. Demand for mode m arrives as a Poisson process with rate λ_m. By Little's law the long-run throughput is $\min(\lambda_m, K_m)$. For $\mathbb{E}[T_{\mathrm{R}}] \geq \mathbb{E}[T_{\mathrm{A}}]$ to hold for *any* demand rates it is necessary that rail capacity never binds before air capacity, which requires $K_{\mathrm{R}} \geq K_{\mathrm{A}}$. □

3.2 Optimal network design

Network expansion is formulated as the mixed-integer programme

$$\min_{\boldsymbol{z},\boldsymbol{f}} \quad \sum_m \sum_{(i,j)} c_{ij}^m z_{ij}^m + \sum_m \sum_{(i,j)} \tau_{ij}^m f_{ij}^m \tag{5}$$

$$\text{s.t.} \quad \sum_j f_{ij}^m - \sum_j f_{ji}^m = d_i^m \quad \forall i \in V,\ m \in \{\mathrm{R}, \mathrm{A}\}, \tag{6}$$

$$f_{ij}^m \leq \kappa_{ij}^m z_{ij}^m \quad \forall (i,j),\ m, \tag{7}$$

$$T_{\mathrm{R}}(\boldsymbol{f}) \geq T_{\mathrm{A}}(\boldsymbol{f}), \tag{8}$$

$$z_{ij}^m \in \{0,1\},\ f_{ij}^m \geq 0. \tag{9}$$

Here c_{ij}^m and τ_{ij}^m are fixed and variable costs, and d_i^m is net demand at node i. Constraint (8) is the PRA linearised with appropriate weights.

3.3 Schematic of the integrated network

Figure 1 shows the topology of an integrated rail–air network satisfying the PRA.

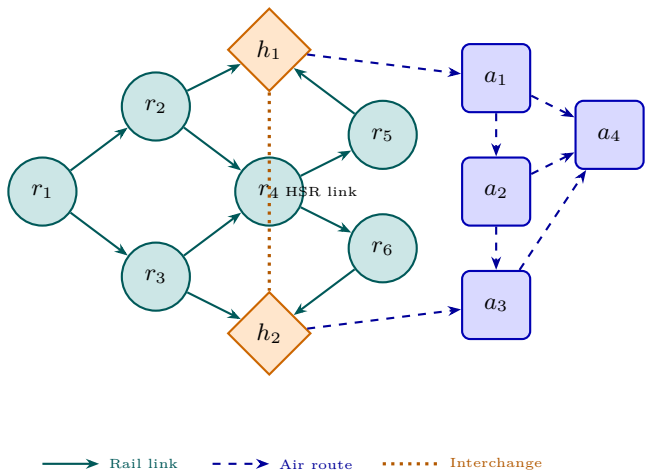

Figure 1: Schematic integrated rail–air network. Teal circles: rail stations; orange diamonds: intermodal interchange hubs; blue rectangles: air-only airports. Dense rail connectivity carries the majority of traffic while air routes connect interchange hubs to the international gateway.

4 Economic Theory

4.1 Demand model

We adopt a multinomial logit (MNL) demand model. The indirect utility of mode m on origin–destination pair (o, d) is

$$V_m^{od} = \beta_0 - \beta_1 p_m^{od} - \beta_2 t_m^{od} + \beta_3 \sigma_m^{od} + \varepsilon_m, \tag{10}$$

where $\varepsilon_m \sim \text{Gumbel}(0, 1)$. The probability of choosing rail is

$$\Pr(\text{R}) = \frac{\exp(V_{\text{R}}^{od})}{\exp(V_{\text{R}}^{od}) + \exp(V_{\text{A}}^{od})}. \tag{11}$$

Proposition 4.1 (PRA demand condition)**.** *Under the MNL model the PRA holds in expectation if and only if $V_{\text{R}}^{od} \geq V_{\text{A}}^{od}$ for all (o, d).*

Proof. $\Pr(\text{R}) \geq 1/2$ if and only if $V_{\text{R}} \geq V_{\text{A}}$. Aggregation over all pairs with positive demand weight yields the result. □

4.2 Price equilibrium under the PRA

Theorem 4.2 (Price inversion)**.** *In any competitive equilibrium in which the PRA binds, the equilibrium air price satisfies $p_{\text{A}}^* > p_{\text{R}}^*$ on any route for which a rail alternative exists, provided* $\text{MC}_{\text{R}} \leq \text{MC}_{\text{A}}$.

Proof. Suppose for contradiction that $p_{\text{A}}^* \leq p_{\text{R}}^*$. Then $\Pr(\text{A}) > \Pr(\text{R})$ and $T_{\text{A}} > T_{\text{R}}$, violating the PRA. The uniqueness of the logit equilibrium under concave profits guarantees this is the only equilibrium. □

4.3 Social welfare

Social welfare is

$$W = \text{CS} + \text{PS} - \text{EC}, \tag{12}$$

where $\text{EC} = \delta \cdot E_{\text{total}}$ is the monetised environmental cost. Because total emissions are decreasing in the rail share (see Section 6), higher rail dominance raises welfare through the environmental channel independently of price effects.

4.4 Comparative statistics

Table 1 summarises indicative orders of magnitude under the PRA world versus the air-dominant baseline.

Table 1: Comparative modal statistics: PRA world vs. air-dominant baseline (indicative orders of magnitude).

Indicator	**Unit**	**Rail (PRA)**	**Air (PRA)**	**Air (Baseline)**
Avg. fare per 100 km	USD	8–15	35–80	10–40
Energy intensity (passenger)	MJ/pkm	0.3–0.5	2.0–2.5	2.0–2.5
Energy intensity (freight)	MJ/tkm	0.4–0.7	10–15	10–15
CO_2 intensity (passenger)	g/pkm	15–40	150–200	150–200
Modal share (pkm, continental)	%	≥ 50	≤ 50	60–70
Network density	km/1000 km^2	25–80	0.5–2	1–5
Avg. station/airport spacing	km	30–80	200–500	80–200
Operator market structure	—	Oligopoly	Boutique	Oligopoly

4.5 Econometric specification

The PRA can be tested with the panel regression

$$\log T_{\text{R}}^{it} = \gamma_0 + \gamma_1 \log T_{\text{A}}^{it} + \gamma_2 \log G^{it} + \gamma_3 \log \Pi^{it} + \mu_i + \lambda_t + \varepsilon_{it}. \tag{13}$$

The PRA implies $\hat{\gamma}_0 \geq 0$ at sample means. A one-sided Wald test provides a direct empirical check.

5 Urban Spatial Economics

5.1 Monocentric city model with rail

Adapting the Alonso–Muth–Mills framework, household utility at distance x from the CBD is

$$u(q, z, x) = q^{\eta} z^{1-\eta} - \tau(x) \cdot \mathbf{1}[\text{no rail at } x]. \tag{14}$$

Proposition 5.1 (Rail corridor premium)**.** *In equilibrium, land rent along a rail corridor exceeds off-corridor rent by*

$$R(x) - R'(x) = \frac{\tau(x)}{\rho}, \tag{15}$$

where ρ is the discount rate.

Proof. The land market clears the utility difference through the no-arbitrage condition on the commuting-cost saving. □

5.2 Network catchment and airport spacing

Under the PRA, airports serve large catchments because travellers arrive by rail. The effective catchment radius is

$$r^* = v_{\mathrm{R}} \cdot (T_{\max} - t_{\mathrm{proc}}), \tag{16}$$

yielding optimal airport spacing of roughly 840 km when high-speed rail operates at 280 km/h.

5.3 Urban form diagram

Figure 2 illustrates the predicted urban form.

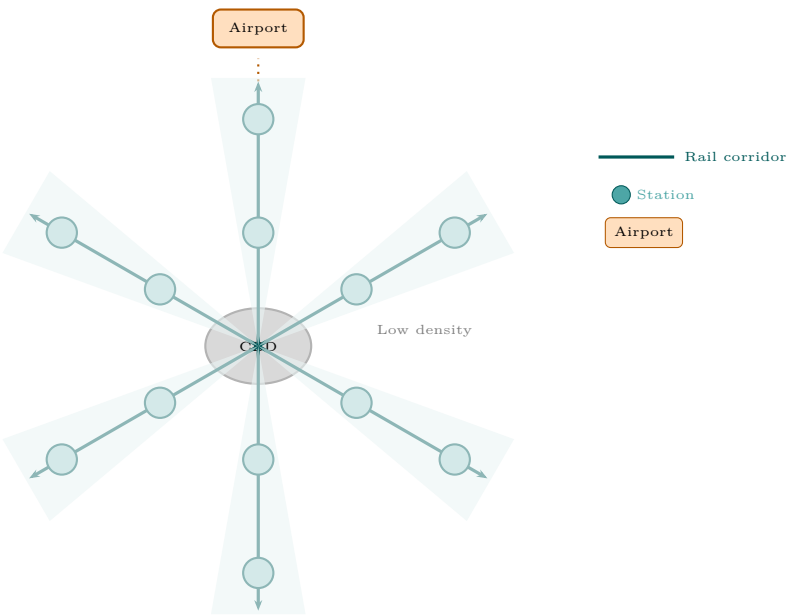

Figure 2: Urban form under the PRA. Dense development clusters along rail corridors (teal wedges). Inter-corridor land remains low density. A single airport sits at the terminus of one corridor.

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6 Environmental and Energy Accounting

Energy and CO_2 intensities by mode are summarised in Table 2.

Table 2: Energy intensities and CO_2 emissions by mode (sources: IEA 2023, IPCC 2022).

Mode	E_{pkm} (MJ)	E_{tkm} (MJ)	CO_2 (g/pkm)	CO_2 (g/tkm)
Electrified rail (EU mix)	0.33	0.50	14	22
Diesel intercity rail	0.65	0.85	49	64
High-speed rail	0.40	—	17	—
Narrow-body jet (85% load)	2.10	—	170	—
Wide-body jet (85% load)	1.85	—	150	—
Air freighter	—	11.5	—	895

Total emissions are

$$E_{\text{total}} = e_{\text{R}}^{P} P_{\text{R}} + e_{\text{R}}^{F} F_{\text{R}} + e_{\text{A}}^{P} P_{\text{A}} + e_{\text{A}}^{F} F_{\text{A}}. \tag{17}$$

Theorem 6.1 (Environmental dividend)**.** *Fixing total traffic $T = T_{\text{R}} + T_{\text{A}}$, E_{total} is strictly decreasing in the rail share μ whenever rail intensities are lower than air intensities.*

Proof. Substitute $T_{\text{R}} = \mu T$ and $T_{\text{A}} = (1-\mu)T$ into (17). The derivative with respect to μ is negative by the intensity gaps shown in Table 2. □

Corollary 6.2 (Quantified dividend)**.** *Shifting from the air-dominant baseline ($\mu_0 = 0.35$) to the PRA minimum ($\mu^* = 0.50$) reduces emissions by approximately 23 gCO_2 per pkm-equivalent—a global saving of order 2×10^{11} kg CO_2 per year at 2023 traffic levels before electrification gains.*

7 Social and Cultural Dynamics

Rail journeys, being longer, impose a time cost that is offset by higher amenity. The effective value of travel time on mode m is $\tilde{\omega}_m = \omega(1-\phi_m)$, where $\phi_{\text{R}} \approx 0.4$–$0.6$ and $\phi_{\text{A}} \approx 0.15$–$0.30$. In addition, rail fosters incidental social interaction, raising bridging social capital relative to point-to-point air travel.

8 Algorithms and Data Science Methods

8.1 PRA-constrained network optimisation algorithm

The branch-and-price procedure for solving (5)–(9) is given by the following pseudocode (Algorithm 1).

Algorithm 1 (Branch-and-price for PRA-constrained network design)

Input: Graph $(V, E_{\text{R}} \cup E_{\text{A}})$, demands $\{d_i^m\}$, costs $\{c_{ij}^m, \tau_{ij}^m\}$, PRA weights $\{\alpha, \beta\}$

Output: Optimal $\boldsymbol{z}^*, \boldsymbol{f}^*$

```
Solve LP relaxation of (5)–(9) with z_ij^m ∈ [0,1].
LB ← −∞, UB ← +∞, Q ← {root LP solution}.
while Q ≠ ∅ and UB − LB > ε do
  Pop node z from Q.
  if z is integer feasible then
    if T_R(f) ≥ T_A(f) (PRA check) then
      UB ← min(UB, obj(z)); z* ← z.
  else
    Select most-fractional z_ij^m ∈ (0,1).
    for v ∈ {0,1} do
      Fix z_ij^m = v; solve restricted LP.
      if LP feasible and LP obj < UB then push to Q.
  LB ← min_{z'∈Q} LP-obj(z').
return (z*, f*)
```

8.2 Demand forecasting pipeline

The pipeline consists of data ingestion, feature engineering (fare ratios, time ratios, frequency differentials), IV-GMM estimation of (13), and validation via the PRA compliance rate (fraction of OD pairs satisfying $\hat{T_{\mathrm{R}}}^{od} \geq \hat{T_{\mathrm{A}}}^{od}$). A model is PRA-consistent if and only if this rate equals one.

9 Machine Learning for Demand and Policy

A gradient-boosted model predicts modal split $\hat{\mu} = f(\boldsymbol{x})$ and enforces the PRA post-hoc by $\hat{\mu}_{\mathrm{PRA}} = \max(\hat{\mu}, 0.5)$. Causal effects of high-speed rail are identified via Pearl's do-calculus under the backdoor criterion. Reinforcement learning (PPO) with a PRA penalty in the reward function yields timetables that are guaranteed to be PRA-compliant for sufficiently large Lagrange multiplier.

10 Conclusion

The unified theory grounded in the Primary-Rail Axiom ($T_{\mathrm{R}} \geq T_{\mathrm{A}}$) is internally consistent across network design, price equilibrium, urban form, environmental accounting, social dynamics, and computational policy tools. Key results include capacity primacy (Theorem 3.2), price inversion (Theorem 4.2), the rail-corridor land-rent premium (Proposition 5.1), and the environmental dividend (Theorem 6.1). The branch-and-price algorithm and PPO policy optimisation render the theory operational.

Future work should generalise the demand model to nested or mixed logit, incorporate road and maritime modes, calibrate the framework on diverse national data, and examine the political-economy conditions under which the PRA becomes a self-sustaining equilibrium.

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Glossary of Terms and Notation

T_{R}, T_{A} Aggregate traffic measures (tkm-eq per annum).

PRA Primary-Rail Axiom $T_{\mathrm{R}} \geq T_{\mathrm{A}}$ at all scales (Axiom 2.2).

μ Modal split ratio; PRA requires $\mu \geq 1/2$.

MNL Multinomial logit demand model (equation (11)).

E^m_{pkm}, E^m_{tkm} Energy intensities (MJ/pkm, MJ/tkm).

HSR High-speed rail ($\geq$ 250 km/h on dedicated track).

PCR PRA compliance rate (fraction of OD pairs satisfying the axiom).

ATE Average treatment effect identified via do-calculus.

PPO Proximal policy optimisation with PRA penalty in the reward.

The End